

Ch. 12 Studyguide

Key Concepts

12.1 The Arithmetic of Equations

- A balanced chemical equation provides the same kind of quantitative information that a recipe does.
- Chemists use balanced chemical equations as a basis to calculate how much reactant is needed or product is formed in a reaction.
- A balanced chemical equation can be interpreted in terms of different quantities, including numbers of atoms, molecules, or moles; mass; and volume.
- Mass and atoms are conserved in every chemical reaction.

12.2 Chemical Calculations

- In chemical calculations, mole ratios are used to convert between moles of reactant and moles of product, between moles of reactants, or between moles of products.
- In a typical stoichiometric problem, the given quantity is first converted to moles. Then the mole ratio from the balanced equation is used to calculate the moles of the wanted substance. Finally, the moles are converted to any other unit of measurement related to the unit mole.

12.3 Limiting Reagent and Percent Yield

- In a chemical reaction, an insufficient quantity of any of the reactants will limit the amount of product that forms.
- The percent yield is a measure of the efficiency of a reaction performed in the laboratory.

Supplemental - Hydrates:

- A hydrate is a compound that contains a specific number of water molecules bound to its atoms.
- In a balanced formula for a hydrate, the number of water molecules is written after a dot following the formula of the substance.
- The water in a hydrate is called the water of hydration; this water can be driven off by heating the crystal to leave behind an anhydrous substance.

- Mass is conserved when a hydrate is heated; the mass of the hydrate is equal to the sum of the mass of the anhydrous salt and the mass of the water vapor released.
- The mole ratio of water to the anhydrous salt is used to determine the specific formula and name of the hydrate.

Vocabulary

- | | |
|--------------------|---------------------|
| ● actual yield | ● percent yield |
| ● excess reagent | ● stoichiometry |
| ● limiting reagent | ● theoretical yield |
| ● mole ratio | |

Key Equations

- **mole-mole relationship for $aG \rightarrow bW$:** $x \text{ mol } G \times (b \text{ mol } W / a \text{ mol } G) = xb/a \text{ mol } W$
- **percent yield:** $\text{percent yield} = (\text{actual yield} / \text{theoretical yield}) \times 100\%$
- Supplemental - Hydrates
 - **Mass of Water Lost:** $\text{Mass of water} = \text{Mass of hydrate} - \text{Mass of anhydrous salt}$
 - **Determining the Hydrate Ratio (x):** $x = \text{moles of } H_2O / \text{moles of anhydrous salt}$
 - **General Formula of a Hydrate:** $(\text{Formula of salt}) \cdot xH_2O$